CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Micro:bit Sensor Station

Goal: Program a classroom micro:bit or confirmed equivalent to detect a condition and produce a useful visual, sound, or motion alert.

35 minutes | 8 steps | Computer Science and Programming, Digital Literacy | Playbook B meeting 29

Ask | Plan | Build | Test | Improve

You are done when

Demonstrate three input tests and explain the sensor, threshold, output, and one revision.

You will use

- Classroom micro:bits or confirmed equivalents
- USB cables
- Computers
- Teacher-approved sensors if available
- Starter code cards
- Sensor data cards
- Paper code blocks
- Test record

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 5 min

- Place the micro:bit beside the computer.
- Open the teacher starter file.
- Connect only the approved cable.
- Wait until the computer shows the board.

2 Step 2: Device safety — 4 min

- Use only approved USB power.
- Connect no unknown device.
- Handle the board by its edges.
- Disconnect power when the teacher says stop.

3 Step 3: Plan the sensor — 5 min

- Choose one classroom condition.
- Write what the board will measure.
- Choose the number that turns the alert on.
- Choose a clear alert.

4 Step 4: Build the program — 6 min

- Place the measure block first.
- Add the number check.
- Add the alert block.
- Transfer the program to the approved device.

5 Step 5: Test three values — 6 min

- Test one number below the turn-on number.
- Test the turn-on number.
- Test one number above it.
- Record each alert.

6 Step 6: Tune the alert — 3 min

- Change the turn-on number or alert.
- Repeat the three tests.

7 Step 7: Disconnect safely — 3 min

- Stop the program.
- Unplug the USB cable by its connector.
- Return the board and cable to the tray.

8 Step 8: Exit check — 3 min

- State the final turn-on number.
- Name one limit of the data.

What to turn in

Three input tests and your sensor, threshold and alert.

No computer? Another way to do it

Use paper code blocks to plan and run the program by hand.

Finished early? Next challenge

Two-Level Alert: Add one warning level before the main threshold so the user can respond earlier.

Stay safe

Use only approved classroom hardware and USB power, connect no unknown devices, collect no names or personal sound recordings, and handle sensors by the demonstrated edges.

Grade targets

Grade 5: Show an alert when a button or sensor changes.
Grade 6: Choose a threshold and explain why.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.